



Syllabus

IE 311 - Operations Research I

Summer 2025-2026

Faculty	Faculty of Engineering and Natural Sciences				
Semester	Summer 2025-2026				
Course Code	IE 311				
Course Name	Operations Research I				
Time/Place	Time	Week Day	Place	Date	
	13:40-16:30	Mon	FENS-L055	Jul 6-Aug 21, 2026	
	14:40-17:30	Tue	FENS-L055	Jul 6-Aug 21, 2026	
Level of Course	Undergraduate				
Credits	SU Credit:3, ECTS:6, Basic:2, Engineering:4				
Prerequisites	(MATH 201 or MATH 212) and ENS 208				
Corequisites	IE 311R				
Course Type	Lecture				
Program Requirements	Programs\Req.Type		Required	Core Elective	Area Elective
	Materials Science and Nano Engineering				✓
	Computer Science and Engineering				✓
	Industrial Engineering		✓		
	Data Science and Analytics			✓	
	Molecular Biology, Genetics and Bioengineering				✓
Instructors	Name	Email	Office Hours		
	Erhun Kundakcioglu	erhun.kundakcioglu@sabanciuniv.edu			
Teaching Assistants					
Course Description	Linear and integer programming formulations; convex analysis; algorithmic design and the simplex method; duality and sensitivity; computer implementations.				
Learning Outcomes	1. Model decision problems into integer or continuous linear programming models.				
	2. Apply primal and dual simplex methods to solve linear programming problems				
	3. Have a basic knowledge of the concept of duality in linear programming and its consequences.				
	4. Interpret solutions using simplex optimality conditions, and perform sensitivity and reoptimization analyses.				
	5. Implement linear and integer programming formulations in a programming language.				
Course Objective	The objective of this course is to study the modeling and solution of decision problems with deterministic parameters using operations research techniques with a particular emphasis on solution algorithms and implementation.				
Sustainable Development Goals (SDGs) Related to This Course	-				
Resources	Operations Research: Applications and Algorithms, 4th edition. W.L. Winston. Thomson/Brooks/Cole, 2004.				
Technology Requirements	Students will need to model, implement and solve linear and integer programs in lectures and recitations. We will use Gurobi solver with Python interface.				
Link to LMS/SUCourse	https://sucourse.sabanciuniv.edu/plus/course/view.php?id=13563				
Assessment Methods	Percentage (%)		Number of assessment methods		
	Final	35%			
	Midterm	40%	2		
	Quiz	15%	5		
	Other	10%			

Assessment Info	Other: There will be 10 lab sessions in total, and together they account for 10% of the final course grade.														
Weekly Content	<table> <tr> <td>Week 1</td><td>Introduction to Optimization, Modeling Linear Programs</td></tr> <tr> <td>Week 2</td><td>Modeling Linear Programs</td></tr> <tr> <td>Week 3</td><td>Modeling Integer Programs</td></tr> <tr> <td>Week 4</td><td>Analysis of Linear Programming, Simplex Method for structured LPs</td></tr> <tr> <td>Week 5</td><td>Simplex Method for unstructured LPs</td></tr> <tr> <td>Week 6</td><td>Duality, Dual Simplex and Revised Simplex Methods</td></tr> <tr> <td>Week 7</td><td>Sensitivity and post-optimality analysis</td></tr> </table>	Week 1	Introduction to Optimization, Modeling Linear Programs	Week 2	Modeling Linear Programs	Week 3	Modeling Integer Programs	Week 4	Analysis of Linear Programming, Simplex Method for structured LPs	Week 5	Simplex Method for unstructured LPs	Week 6	Duality, Dual Simplex and Revised Simplex Methods	Week 7	Sensitivity and post-optimality analysis
Week 1	Introduction to Optimization, Modeling Linear Programs														
Week 2	Modeling Linear Programs														
Week 3	Modeling Integer Programs														
Week 4	Analysis of Linear Programming, Simplex Method for structured LPs														
Week 5	Simplex Method for unstructured LPs														
Week 6	Duality, Dual Simplex and Revised Simplex Methods														
Week 7	Sensitivity and post-optimality analysis														
Course Policies	AI/LLM Usage Policy All coursework in this class consists of lab and recitation assignments that must be completed during class sessions. Students are not permitted to use AI-based tools (such as ChatGPT, Claude, Bard, Grammarly, or similar platforms) for any part of these graded assignments. This prohibition includes -but is not limited to- brainstorming, outlining, drafting, editing, paraphrasing, or generating solutions. All submitted work must reflect each student's own independent effort and original thought. This restriction applies uniformly across all lab and recitation work. Use of AI tools during class for any graded task is strictly prohibited. Outside of class, students are welcome to explore AI tools to aid their understanding of course concepts. However, please be aware that these tools may produce inaccurate or incomplete explanations, particularly for mathematical optimization problems. Students remain fully responsible for mastering the material and verifying their understanding. Lecture Policy • First two classes of each week will be theoretical lectures. • Third class of each week will be computer-based labs and there will be implementation tasks to be performed in groups. • In the summer semester that three class hour routine will be repeated twice a week. Lab Policy Groups of up to 5 students will work together on implementation tasks during lab sessions and must submit their answer (typically a Python file) at the end. Students may form their own groups, and each group should bring at least one laptop. One member will serve as the lead student each session, responsible for writing the code with assistance from the others and submitting it on SuCourse+. The lead student role must rotate weekly. Consequently, each student must take the lead and submit group work at least twice during the semester. <u>Failure to do so will result in zero lab grade.</u> Learning Assistants (LAs) will be available to support with mini-questions and technical issues, but they will not complete the work for students. Answers are graded on a 0-1 scale, based on evidence of group effort rather than correctness alone. Attendance alone is not sufficient; students must actively contribute to group work to earn the grade. There will be 10 lab sessions in total. Recitation Policy • The TAs will hold interactive recitation hours. • Learning Assistants (LAs) will be available to support with mini-questions and technical issues, but they will not complete the work for students. • The students are expected to have already worked on the material, studied questions posted, and come prepared to these sessions with their computers. • Quizzes will be held during recitation hours. Quiz Policy • Quizzes will be announced in advance. • Quizzes will be given during recitation hours towards the end of class. • Quizzes will be in-person, individual work and closed-book. • Quizzes will primarily be based on recent topics. • There will be 7 quizzes, and the lowest 2 grades will be dropped. • There will be no make-up quiz whatsoever. Exam Policy • There will be two midterm exams and a final, all in-person and closed-book. • A make-up exam will be given for students missing any of these exams due to a medical excuse at the end of the semester. • Tentative exam topics are as follows: ◦ Midterm 1: Modeling ◦ Midterm 2: Analysis of Linear Programming (up to and including duality) ◦ Final: Comprehensive														
AI Policies	Guidelines on the "Use of Generative AI" for students														

SU Academic Integrity Statement	https://www.sabanciuniv.edu/en/academic-integrity-statement
Health Report Requirements	Student Medical Reports Instruction Letter

SUCourse view | 2026-06-22 17:31:22 +03:00